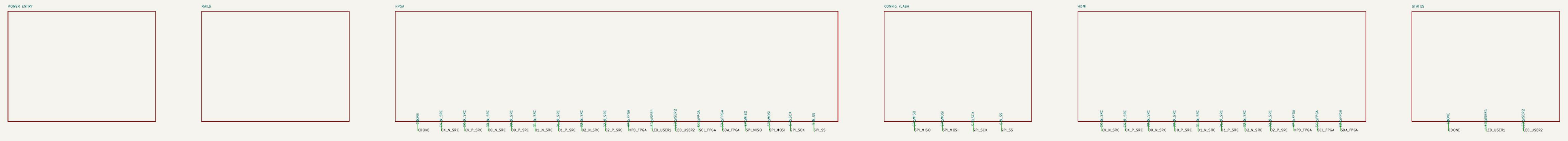
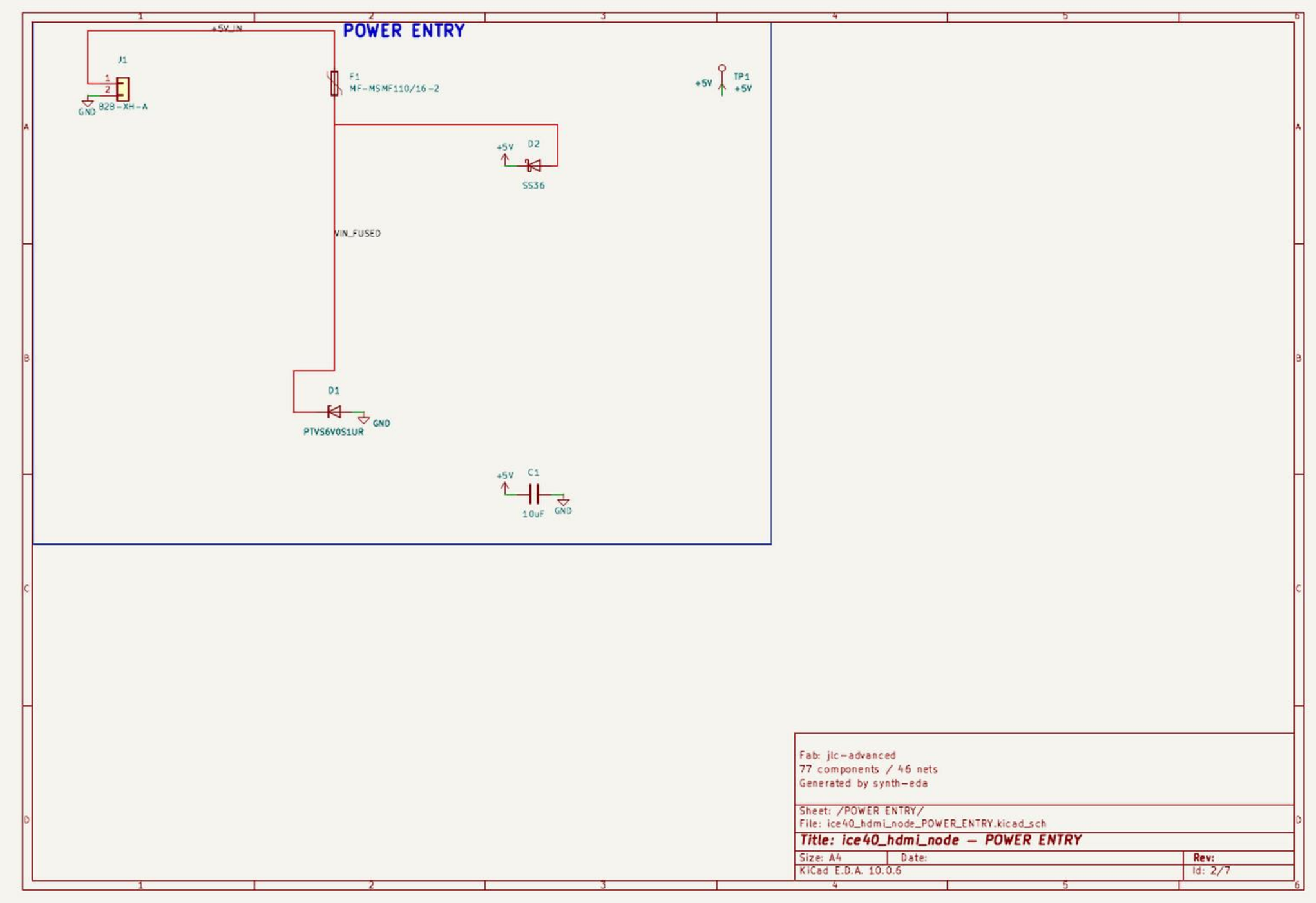
5 V in; PTC 1.1 A hold; TVS 6.0 V standoff above the 5.5 V maximum input.

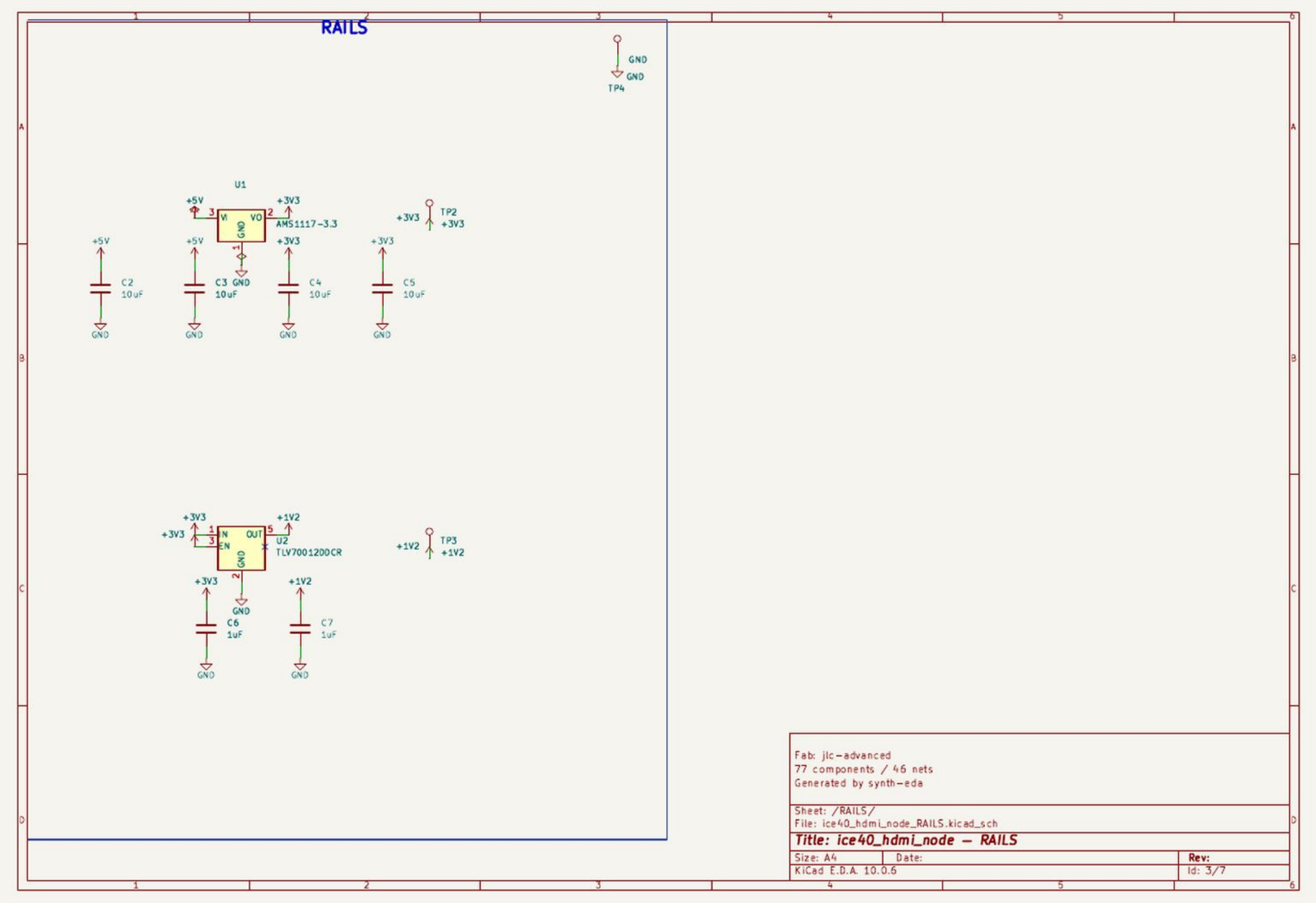
TMDS drive: FPGA LVCMOS33 output + 270 ohm into 50 ohm sink pull-up to 3.3 V.


POWER ENTRY
RAILS
FPGA
CDONE
CK_N_SRC
CK_P_SRC
D0_N_SRC
D0_P_SRC
D1_N_SRC
D1_P_SRC
D2_N_SRC
D2_P_SRC
HPD_FPGA
LED_USER1
LED_USER2
SCL_FPGA
SDA_FPGA
SPI_MISO
SPI_MOSI
SPI_SCK
SPI_SS
CONFIG FLASH
SPI_MISO
SPI_MOSI
SPI_SCK
SPI_SS
HDMI
CK_N_SRC
CK_P_SRC
D0_N_SRC
D0_P_SRC
D1_N_SRC
D1_P_SRC
D2_N_SRC
D2_P_SRC
HPD_FPGA
SCL_FPGA
SDA_FPGA
STATUS
CDONE
LED_USER1
LED_USER2

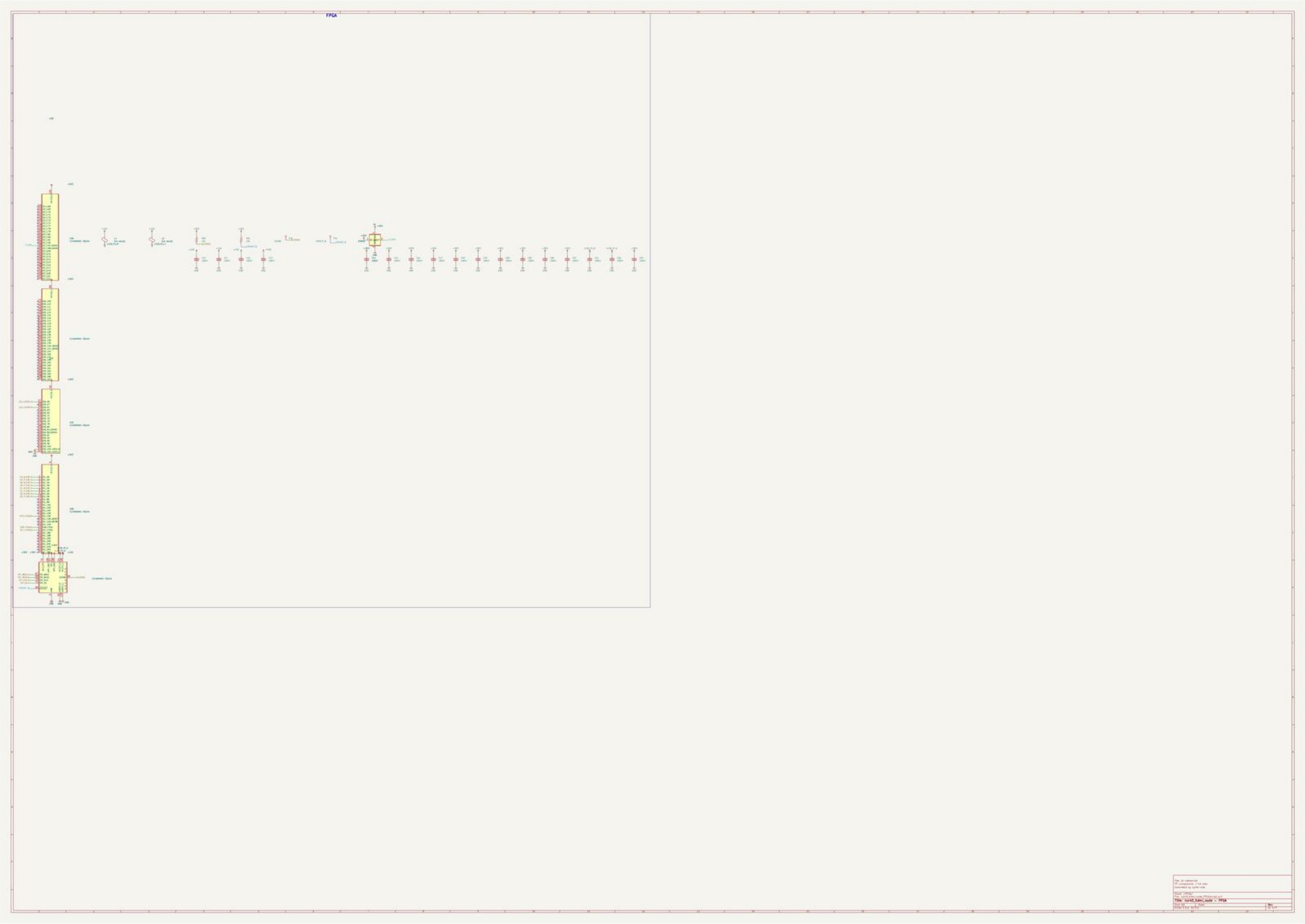
Fab: jlc-advanced
77 components / 46 nets
Generated by synth-eda
Sheet: /
File: ice40_hdmi_node.kicad_sch
Title: ice40_hdmi_node
Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 1/7

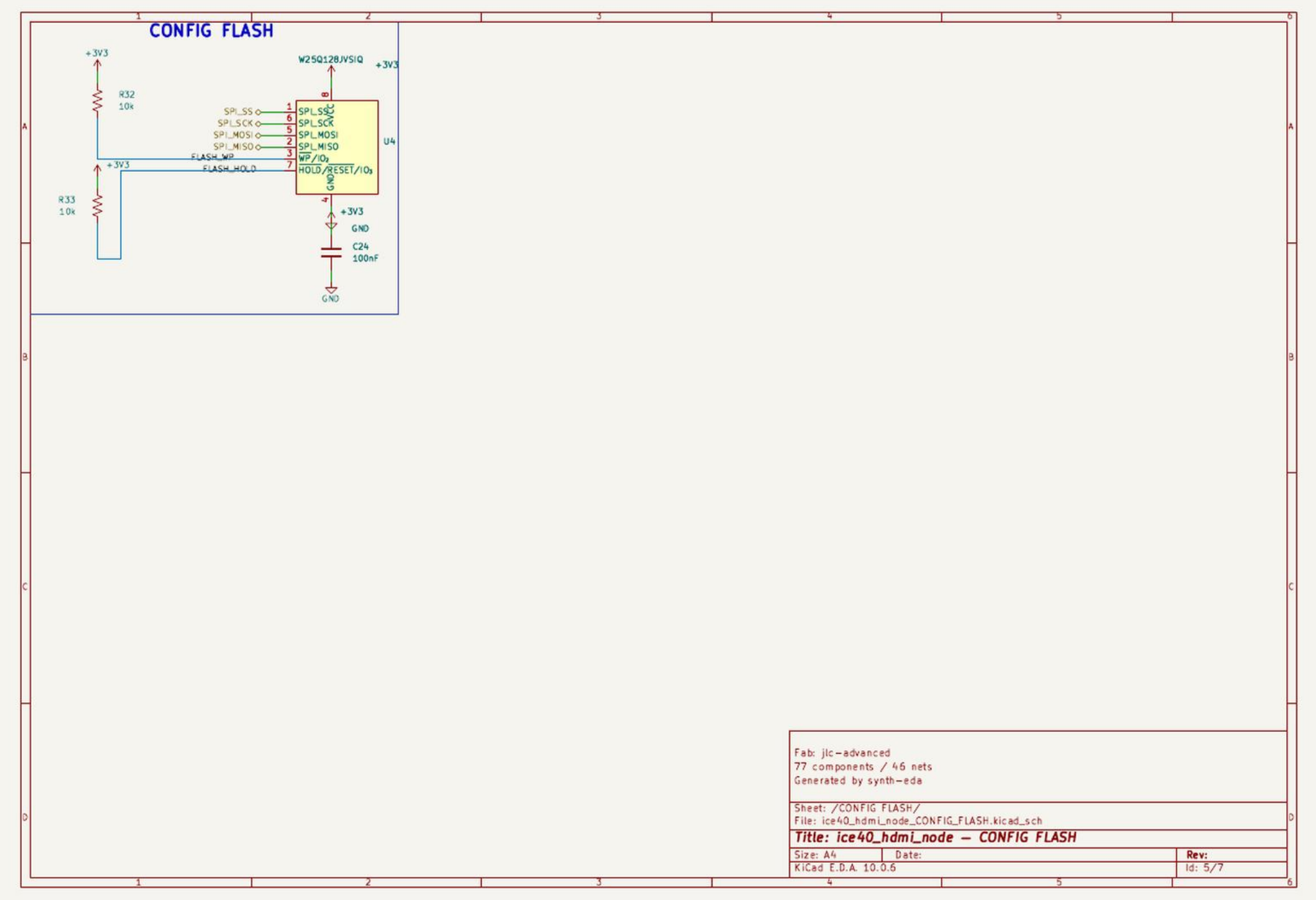

POWER ENTRY
+5V_IN
J1
B2B-XH-A
GND
F1
MF-MSMF110/16-2
+5V
TP1
+5V
D2
SS36
VIN_FUSED
D1
PTVS6V0S1UR
GND
C1
10uF

Fab: jlc-advanced
77 components / 46 nets
Generated by synth-eda
Sheet: /POWER ENTRY/
File: ice40_hdmi_node_POWER_ENTRY.kicad_sch
Title: ice40_hdmi_node — POWER ENTRY
Size: A4 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 2/7

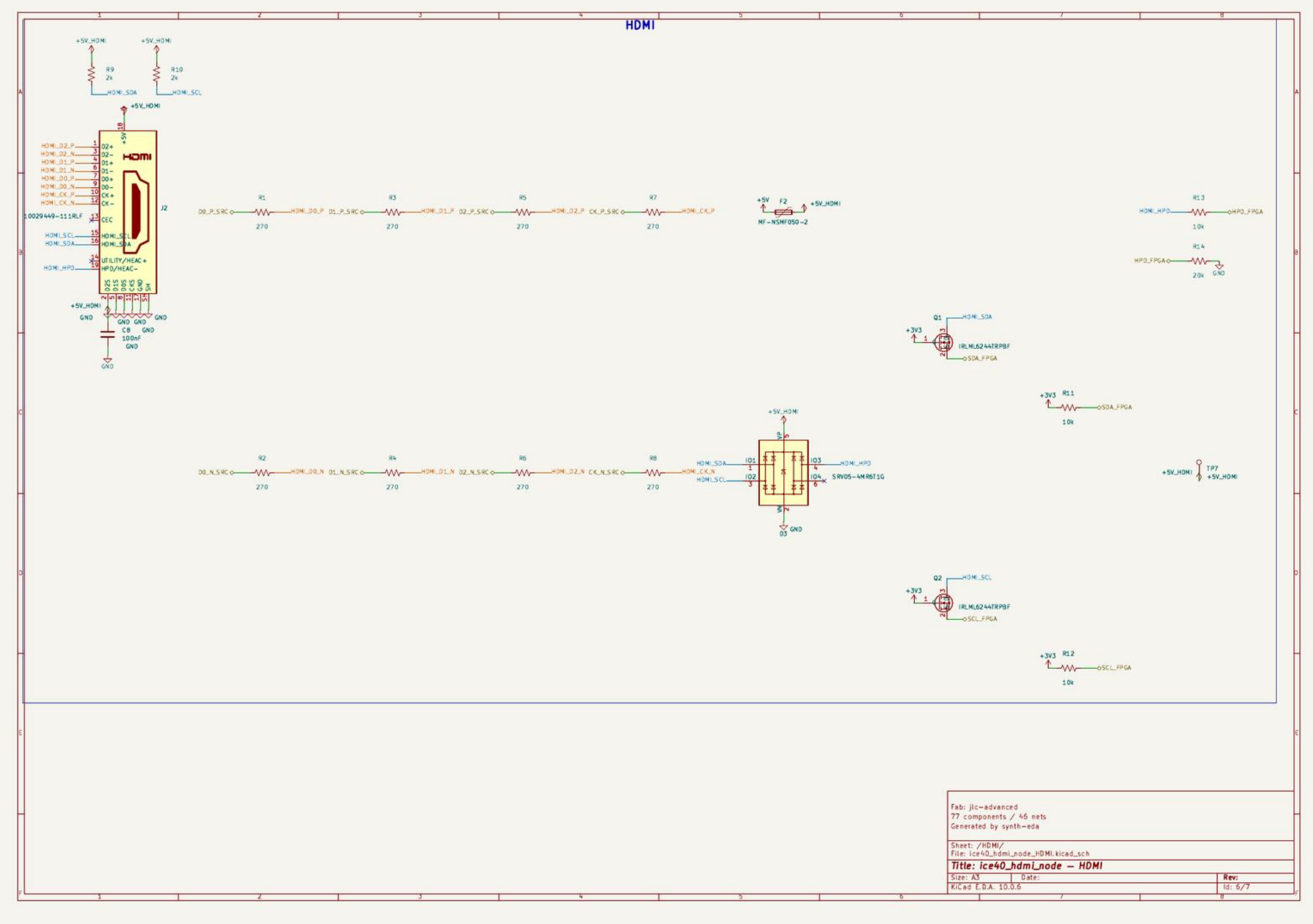

RAILS
GND
TP4
U1
+5V
+3V3
AMS1117-3.3
TP2
C2
10uF
C3
10uF
C4
10uF
C5
10uF
U2
TLV70012DDCR
+1V2
TP3
C6
1uF
C7
1uF

Fab: jlc-advanced
77 components / 46 nets
Generated by synth-eda
Sheet: /RAILS/
File: ice40_hdmi_node_RAILS.kicad_sch
Title: ice40_hdmi_node — RAILS
Size: A4 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 3/7

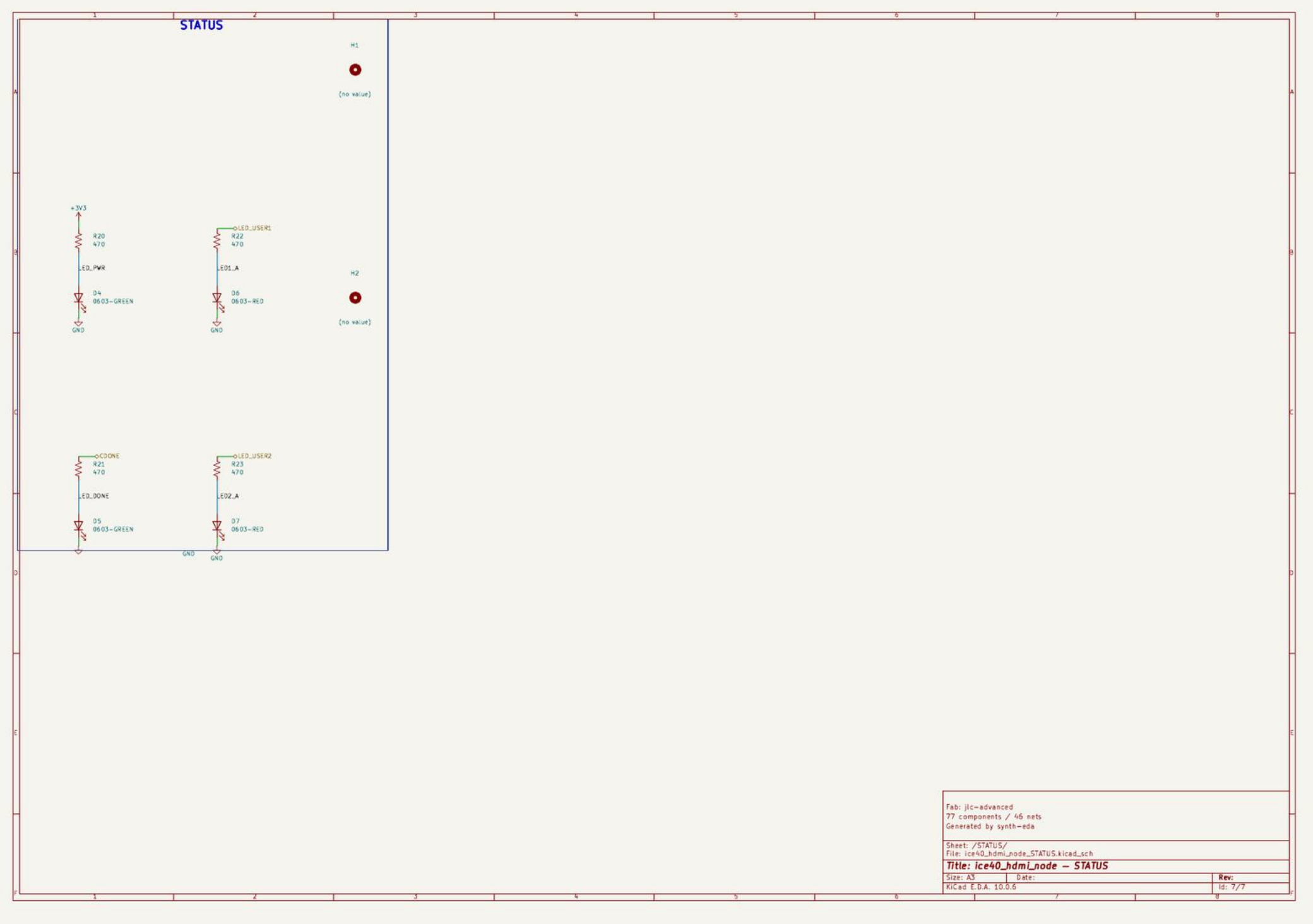



CONFIG FLASH
+3V3
W25Q128JVSIQ
R32
10k
SPI_SS
SPI_SCK
SPI_MOSI
SPI_MISO
WP/IO₂
HOLD/RESET/IO₃
FLASH_WP
FLASH_HOLD
U4
R33
10k
GND
C24
100nF

Fab: jlc-advanced
77 components / 46 nets
Generated by synth-eda
Sheet: /CONFIG FLASH/
File: ice40_hdmi_node_CONFIG_FLASH.kicad_sch
Title: ice40_hdmi_node — CONFIG FLASH
Size: A4 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 5/7


HDMI
+5V_HDMI
R9
2k
R10
2k
HDMI_SDA
HDMI_SCL
J2
10029449-111RLF
HDMI_D2_P
HDMI_D2_N
HDMI_D1_P
HDMI_D1_N
HDMI_D0_P
HDMI_D0_N
HDMI_CK_P
HDMI_CK_N
HDMI_HPD
CEC
UTILITY/HEAC+
HPD/HEAC-
C8
100nF
GND
R1
R3
R5
R7
R2
R4
R6
R8
270
D0_P_SRC
D1_P_SRC
D2_P_SRC
CK_P_SRC
D0_N_SRC
D1_N_SRC
D2_N_SRC
CK_N_SRC
+5V
F2
MF-NSMF050-2
D3
SRV05-4MR6T1G
Q1
Q2
IRLML6244TRPBF
SDA_FPGA
SCL_FPGA
+3V3
R11
10k
R12
10k
R13
10k
R14
20k
HPD_FPGA
TP7

Fab: jlc-advanced
77 components / 46 nets
Generated by synth-eda
Sheet: /HDMI/
File: ice40_hdmi_node_HDMI.kicad_sch
Title: ice40_hdmi_node — HDMI
Size: A3 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 6/7


STATUS
H1
(no value)
H2
(no value)
+3V3
R20
470
LED_PWR
D4
0603-GREEN
LED_USER1
R22
470
LED1_A
D6
0603-RED
CDONE
R21
470
LED_DONE
D5
0603-GREEN
LED_USER2
R23
470
LED2_A
D7
0603-RED
GND

Fab: jlc-advanced
77 components / 46 nets
Generated by synth-eda
Sheet: /STATUS/
File: ice40_hdmi_node_STATUS.kicad_sch
Title: ice40_hdmi_node — STATUS
Size: A3 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 7/7